Introduction to Embedded Systems
Spring 2026
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Version 1.1
Latest Revision: 05/04/26

PCB Design Considerations and Decisions

Link to Altium Project:

[GasControlValve - Emirhan Kir]

Table of Changes

Version	Revisions and Changes
1.0	Initial Document
1.1	Added Altium project link, top-level schematic, top-layer view, bottom-layer view, and 3D view of PCB.
1.2	Added PCB design decisions and updates/changes to the PCB.

1. PCB Overview

1.1 Top-Level Hierarchical View of PCB Schematic

Figure 1:

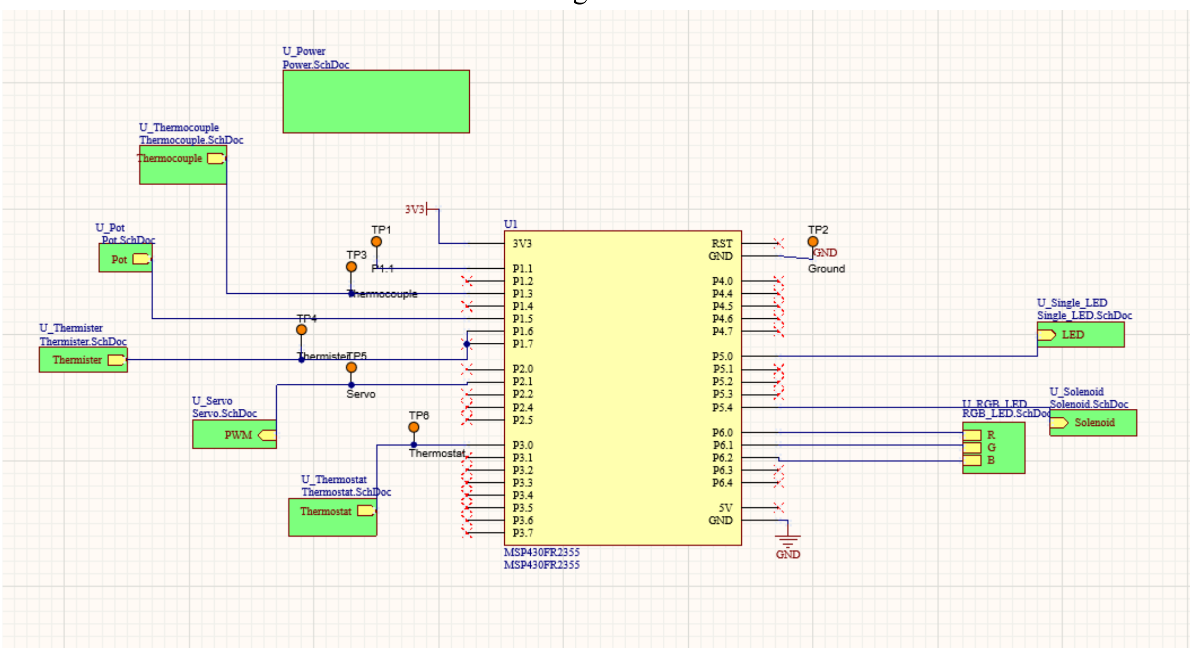
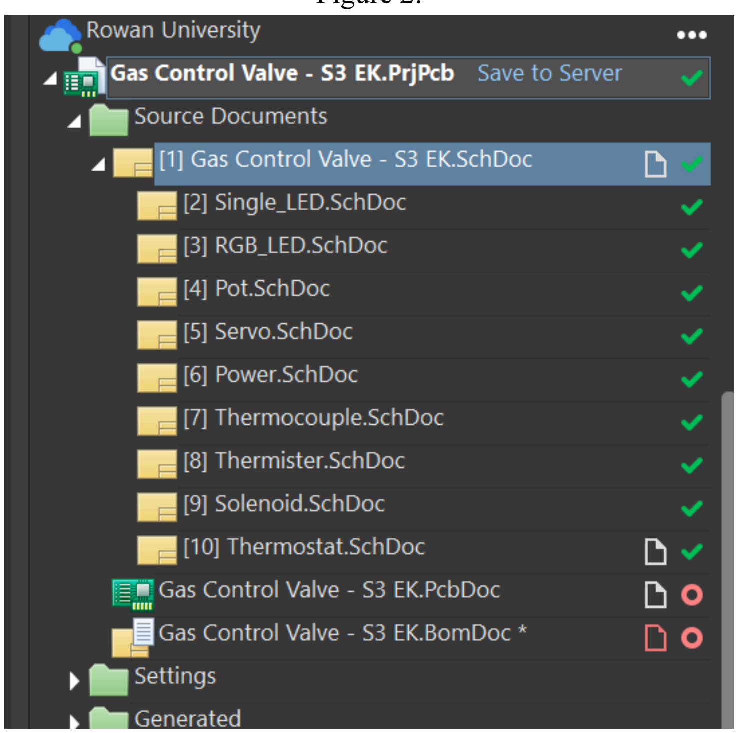
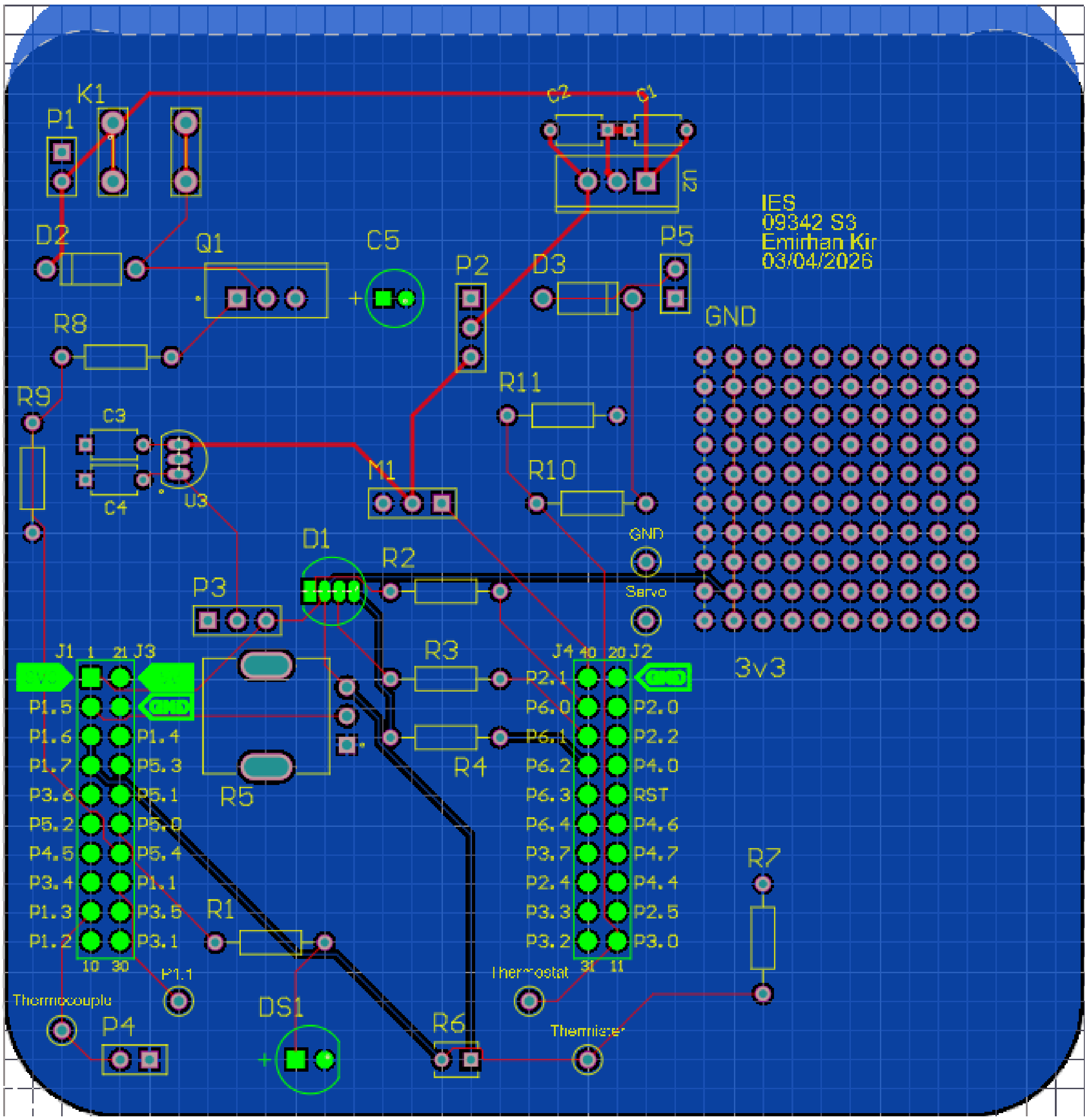


Figure 2:



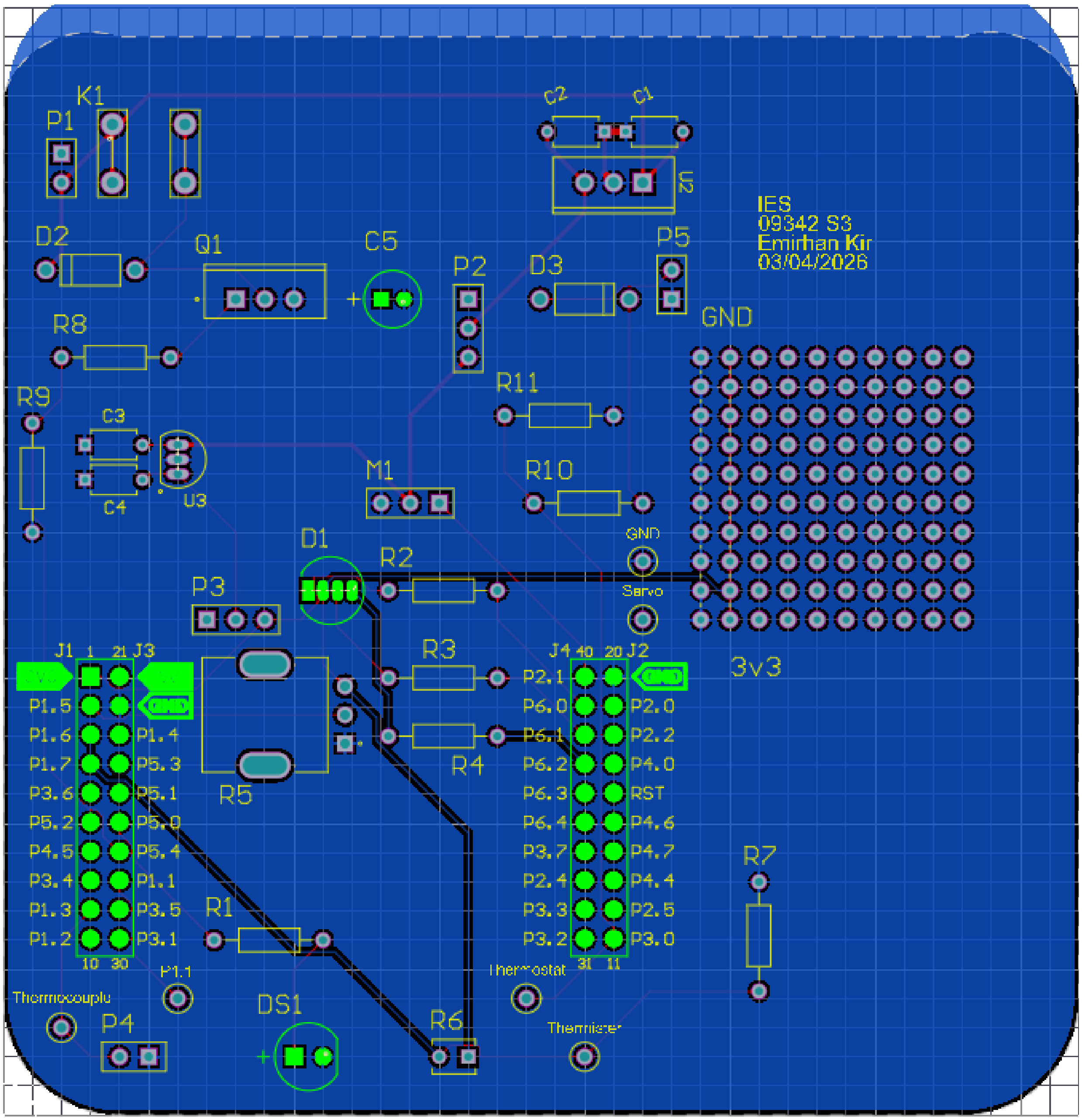
1.2 Top-Layer View of Board

Figure 3:



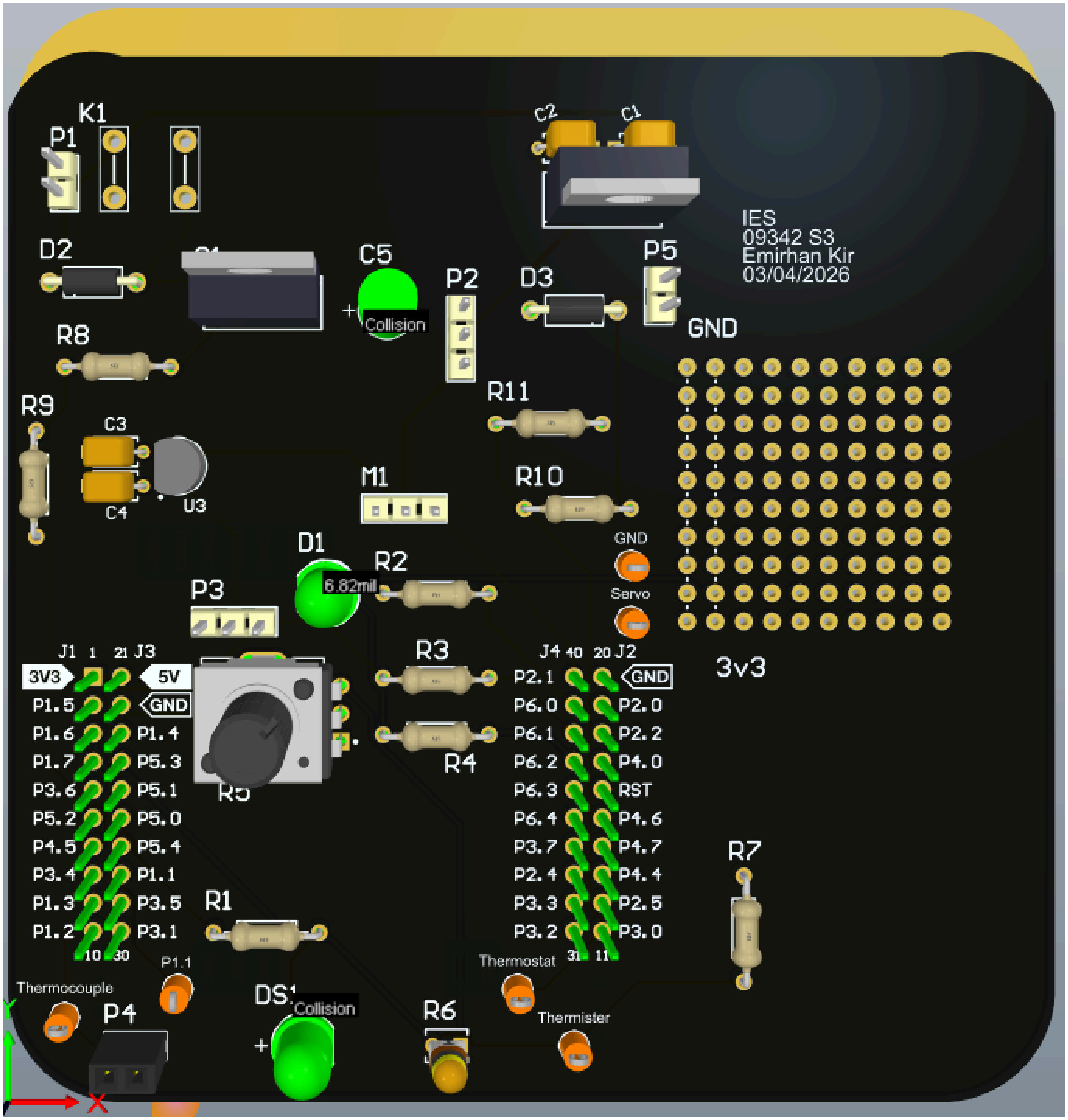
1.3 Bottom-Layer View of Board

Figure 4:



1.4 3D-View of Top of Board

Figure 5:



2. PCB Design Decisions (3-5 components/decisions)

- The thermocouple and thermistor connections were placed near the edge of the board.
  - This made them easier to access during testing since both sensors needed to be checked while the system was running.
- The RGB LED was placed near the P6.0, P6.1, and P6.2 pins.
  - This kept the RGB traces shorter since those pins were used for the LED outputs.
- The solenoid circuit was placed on the left side of the board near the MOSFET and diode.
  - This helped keep the solenoid control parts grouped together and made the valve output easier to debug.
- The servo connector M1 was placed near the middle of the board.
  - This made the servo PWM connection to P2.1 easier to route and kept the servo output close to the MSP430 header.
- Important labels like GND, 3V3, Servo, Thermistor, and Thermocouple were added on the silkscreen.
  - This made the board easier to test and helped prevent wiring mistakes.

3. Updates and Changes to your PCB

- Group related parts closer together.
    - The board works, but some parts that connect together are spread out. In a future version, I would group the input sensors, output devices, power parts, and test points into cleaner sections. This would make the PCB easier to route, debug, and understand when looking at the board later.
- Remove the accidental trace between P1.6 and P1.7.
    - This caused issues when UART needed to be tested because UART required those pins. Removing this trace would let the UART connection be tested normally instead of through back channels.
- Reduce the long diagonal traces across the board.
    - Some traces run across large sections of the PCB, especially from the left header area toward the middle and bottom of the board. In a future revision, I would move parts or reroute traces so the paths are shorter and easier to follow.

References

N/A

Acknowledgements

{1} Russell Trafford: For guiding the project throughout the semester and helping us understand the concepts needed for the design.

{2} Karl Dyer: For explaining the original project goals and helping introduce the schematic design process in Altium.

{3} Michelle Frolio: For giving support with PCB design, Altium, and board building techniques.